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Supercomputing with a stack of PlayStations

By Matt Greenop
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American scientists are taking advantage of the PlayStation3 Cell processor to make low-cost supercomputers.

There's no doubt the latest generation game consoles are packing some serious processing power - but American academics are pushing the PlayStation envelope and used them to build supercomputers.

Dr Gaurav Khanna, a faculty member at the University of Massachusetts who works in the field of theoretical and computational astrophysics, has strung together eight PlayStation3 consoles to make a super-fast cluster to crunch numbers in the name of science.

Khanna says that it's hard to beat the value of his PS3 'gravity grid', which is made possible by the groundbreaking Cell processor technology developed by Sony, IBM and Toshiba.

The processor has a core CPU and six SPUs (Synergistic Processor Units) each of which performs vector operations.

Khanna says that once the PS3 specification was released, he began formulating the plan to build the cluster - despite the fact that his family and friends thought he was joking.

"Sony had been pretty open about its plans for the PS3," he told nzherald.co.nz, "so I had this possibility in mind years before the PS3 was released.

"Indeed, the two main attractive features are that the PS3 is an open platform. Sony does not entirely control what can run on it, which allows one to run scientific code on it - and that it has the revolutionary Cell processor."

The performance that can be milked out the PS3 clusters varies from system to system.

"It depends on what you are running on it and how well it is optimised for the Cell architecture," he explains.

"On my codes, each PS3 is comparable to about 25 processors of an IBM Blue Gene supercomputer. I have heard even higher numbers for other applications. Therefore, the performance of the system is quite impressive"

Khanna's system runs Yellow Dog Linux, chosen because there is a PS3-centric version.

"There are several options for the flavour of PowerPC Linux that one can chose to run on a PS3," he said.

"YDL is particularly well suited because TerraSoft Solutions created a version that is specifically for the PS3. The installation is well documented and very straightforward. You get all you need in one place and don't have to struggle with the somewhat obscure information on various websites."

North Carolina University's Dr Frank Mueller was a pioneer in PS3-based scientific computing.

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